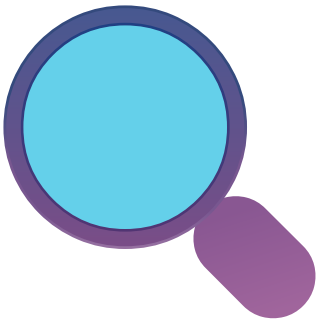
GEOLOGY 101 · CENTRE 1



Look Close!

What can you see on your rock?

1



Pick a rock

2



Look through the glass

3



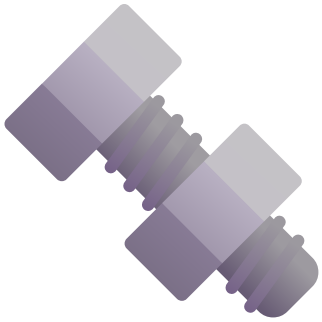
Tell a friend what you see!

👩‍🏫 Helper — what to say and ask

Show them how to hold the magnifying glass **near the rock** and move it slowly until things look big. Ask: **"What colours can you see?" "Are there sparkly bits?" "Any little holes or lines?"** Say: *"God put tiny details in every rock — even the parts we can't see without a magnifying glass!"*

⚠️ **Safety:** never point a magnifying glass at the sun or at a person's eyes. Rocks stay out of mouths and noses.

GEOLOGY 101 · CENTRE 2



Scratch Test

Is your rock HARD or SOFT?

1




A grown-up scratches the rock

2



Look — is there a mark?

3





No mark = HARD
Mark = SOFT

👩‍🏫 Helper — what to say and ask

Scratch each rock firmly with the nail and let the children look at it, and feel it if they want to. Soft rocks like chalk, sandstone and soapstone show a line or leave powder; hard rocks like quartz and granite don't. Ask: **"Did it scratch? So is it hard or soft?"** Have them make two piles: **HARD** and **SOFT**. Say: "*God made strong, hard rocks for building, and soft ones we can even draw with!*" (Bonus: a piece of sidewalk chalk is a soft rock-like thing — let them draw with it.)

⚠️ **Safety:** an **ADULT holds the nail** and does the scratching. Children never hold the nail. A coin or steel spoon works too and is safer.

GEOLOGY 101 · CENTRE 3



Heavy or Light?

Same size… but which one is heavier?

1



Hold one rock in each hand

2



Guess: which is heavier?

3



Put them on the scale to check!

👩‍🏫 Helper — what to say and ask

Use pairs of rocks that are **about the same size**, but one dense (granite, river stone) and one light (lava rock, pumice, chalk). Let them be "human scales" first, with arms out, and hands going up and down. Then weigh each rock on the kitchen scale and point to the bigger number: **"This one is heavier! Did you guess right?"** Say: "*They look the same size, but God made some rocks heavy and some light.*"

⚠️ **Safety:** keep heavy rocks low on the table, not held up high. Feet away from the edge in case one drops.

GEOLOGY 101 · CENTRE 4



Sink or Float?

Will your rock SINK… or FLOAT?

1



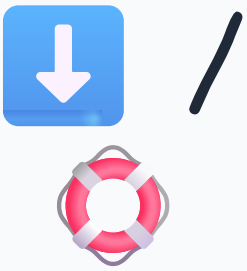
Guess! Thumbs up = float

2



Gently drop it in the water

3

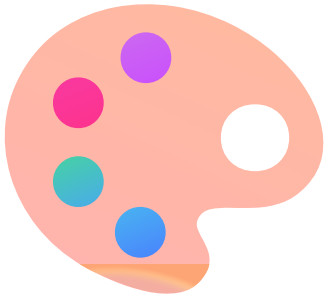


Did it sink or float?

Helper — what to say and ask

Before each drop, everyone guesses with thumbs: **up = float, down = sink**. Most rocks sink, so save the **lava rock / pumice** for last. It floats, and they will be amazed. Ask: **"Why do you think this one floats?"** (It's full of tiny air holes, like a sponge. Let them look at the holes with a magnifying glass.) Say: "*God made a rock that can float! God's world is full of surprises.*"

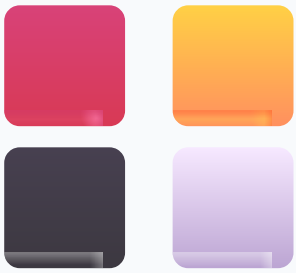
⚠ **Safety:** towel under the bowl. Drop gently: no throwing or splashing. Wipe up spills right away so no one slips.



Sort It!

Sort by COLOUR or by SIZE!

1



Put each rock on its matching colour paper

2



Or drop it in the hole it fits

3



Count how many in each pile!

👩‍🏫 Helper — what to say and ask

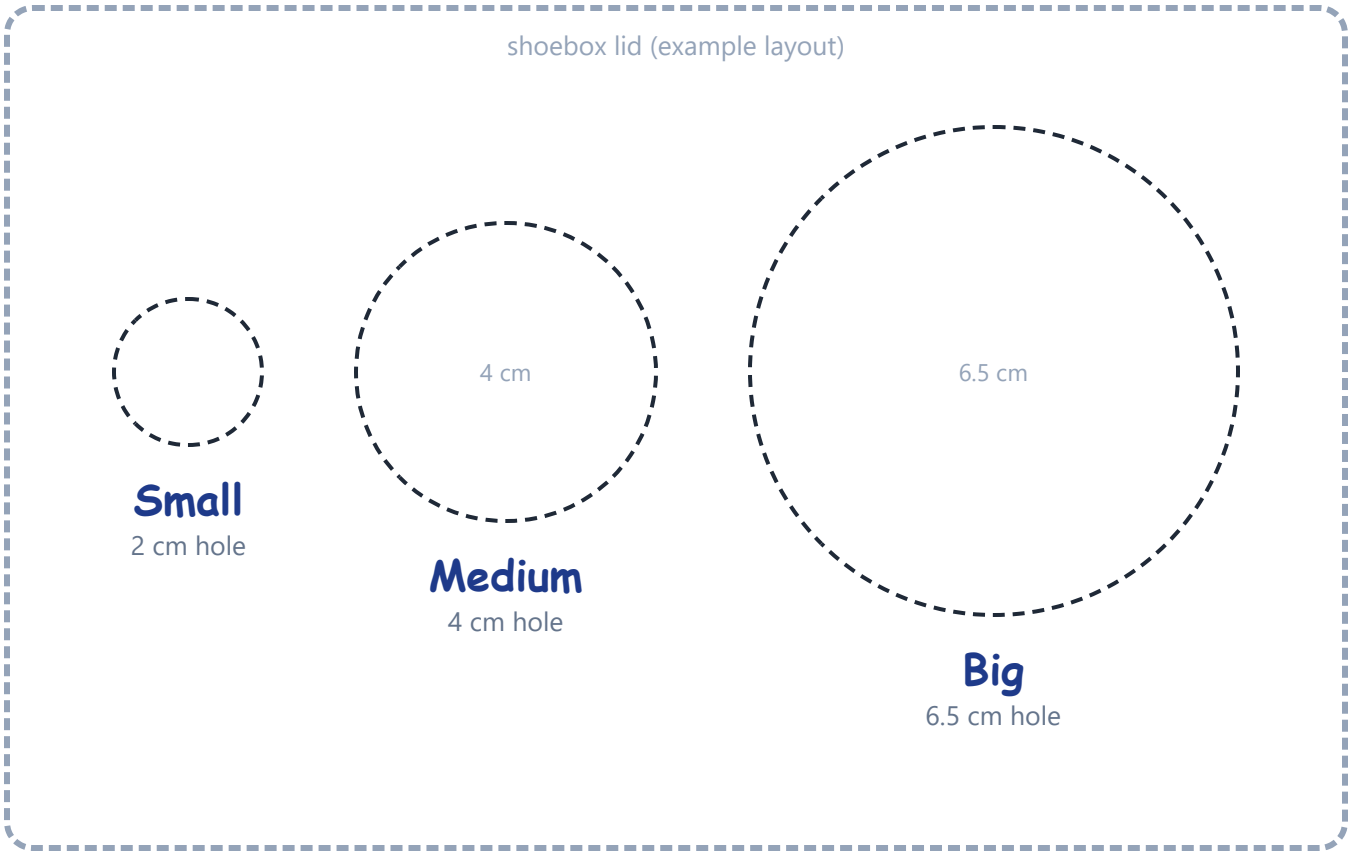
Colour sort: lay out sheets of construction paper (red, grey, white, black, brown, pink) and put each rock on the colour it matches best. **Size sort:** a shoebox lid with three holes cut out (small, medium, big; use the template on the last page). Rocks go through the **smallest hole they fit**. Try small first! Ask: **"Which pile has the most?"** **"Which hole did this one fit?"** Say: "*God made rocks in every colour and every size!*"

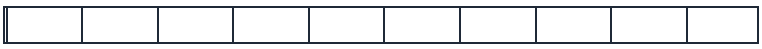
⚠️ **Safety:** pebbles and gravel are small. Watch that they stay out of mouths, noses and ears.

GEOLOGY 101 · CENTRE 5 · SORT IT!

✂️ Shoebox-Lid Size-Sort Template

Cut out these three circles and trace them onto a shoebox lid. An adult cuts the holes with a craft knife. **Print at Scale 100%, Margins None**: the ruler below must measure exactly 10 cm. If it doesn't, the holes are the wrong size.





↑ 10 cm check ruler (each mark = 1 cm)

How to play: a child tries the **small** hole first. If the rock is too big, they try medium, then big. The rock drops into the box. Afterwards, open the lid and count how many rocks went through each hole.

Tip: put a strip of tape round each hole so the cardboard edges stay smooth.